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Course Name: Pattern Recognition1

Assignment1

Exploratory Data Analysis (EDA) and Statistical Hypothesis Testing on the Palmer Penguins Dataset

- **Dataset Description and Preparation**

The dataset consists of 344 observations and 7 features: species, island, bill length, bill depth, flipper length, body mass, and sex.

Initial data inspection revealed missing values in several numerical and categorical variables. To ensure data quality and statistical validity, all rows containing missing values were removed, resulting in a final dataset of 333 complete observations.

Categorical variables such as species, island, and sex were converted into categorical data types to improve computational efficiency and ensure proper handling in analysis.

- **Exploratory Data Analysis (EDA)**

- **Descriptive Statistics**

Descriptive statistics indicate clear variation among penguin species. The average flipper length is approximately 201 mm, while the average body mass is around 4207 grams. The variability in the data suggests meaningful biological differences between species.

These initial observations indicate that differences between species are likely not random.

- **Distribution Analysis**

The distribution of numerical variables shows an approximately normal shape for most features. Both body mass and flipper length exhibit noticeable variability, reflecting differences in penguin size and morphology.

- **Pairwise Relationships (Pairplot)**

The pairplot reveals clear separability among the three penguin species. Gentoo penguins stand out with significantly higher body mass and flipper length values compared to Adelie and Chinstrap species. Adelie and Chinstrap show partial overlap, indicating some similarity in morphological traits.

- **Boxplot Analysis**

Boxplot analysis confirms clear differences among species:

- Gentoo penguins exhibit the highest median values for flipper length and body mass.
- Adelie penguins are generally the smallest.
- Chinstrap penguins fall between the other two species.

These results indicate structural biological differences rather than random variation.

- **Violin Plot Analysis**

The violin plot shows the distribution density of flipper length across species. Gentoo penguins exhibit a concentrated distribution at higher values, indicating relatively consistent large body size. Adelie and Chinstrap show lower and more dispersed distributions.

- **Correlation Analysis**

The correlation analysis reveals a strong positive relationship between body mass and flipper length. This indicates that larger penguins tend to have longer flippers, suggesting a biologically meaningful relationship between body size and morphology.

- **Hypothesis Testing**
- **One-Way ANOVA Test**

Hypotheses:

- H0: There is no significant difference in mean flipper length among species.
- H1: At least one species has a different mean flipper length.

Results:

- Adelie mean flipper length ≈ 190.10 mm
- Chinstrap mean flipper length ≈ 195.82 mm
- Gentoo mean flipper length ≈ 217.24 mm
- F-statistic = 567.41
- p-value $\approx 1.59 \times 10^{-107}$

Statistical Interpretation:

The extremely high F-statistic indicates that variability between species is much larger than variability within species. The p-value is far below the significance

level (0.05), meaning the probability of observing such differences by chance is virtually zero.

Decision:

The null hypothesis is rejected, confirming statistically significant differences in mean flipper length among penguin species

- **Chi-Square Test of Independence**

Hypotheses:

- H0: Species and body mass category are independent.
- H1: Species and body mass category are associated.

Results:

- Chi-Square statistic = 238.80
- p-value = 1.68×10^{-50}

Statistical Interpretation:

The high Chi-Square value indicates a strong discrepancy between observed and expected distributions under independence. The extremely small p-value confirms that this relationship is statistically significant.

Decision:

The null hypothesis is rejected, indicating a strong association between penguin species and body mass category.

- **Discussion**

The combined statistical and visual analyses consistently demonstrate clear biological differences among penguin species. Gentoo penguins are significantly larger in both body mass and flipper length, while Adelie penguins are the smallest. Chinstrap penguins exhibit intermediate characteristics.

The strong positive correlation between body mass and flipper length further supports the biological relationship between size and morphology. Both ANOVA and Chi-Square tests provide strong statistical evidence supporting these observations.

- **Conclusion**

This study confirms that penguin species differ significantly in their biological characteristics. One-Way ANOVA demonstrated significant differences in mean flipper length among species, while the Chi-Square test confirmed a strong association between species and body mass category.

Overall, the analysis highlights the importance of statistical methods in understanding biological data and uncovering meaningful patterns in real-world datasets.